

FUNCTIONAL INEQUALITIES AND SYNCHRONIZATION FOR THE MEASURE-VALUED SOLUTIONS OF STOCHASTIC DIFFERENTIAL EQUATIONS WITH INTERACTION

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ABSTRACT. We study the measure-valued solution to the stochastic differential equations with interaction. We derive pathwise transfer estimates for logarithmic Sobolev, Poincaré, concentration, and T_2 inequalities. The resulting constants are controlled by the maximal Jacobian norm or the Lipschitz distortion of the flow on the support of the initial measure. Under global spatial regularity assumptions, we obtain moment estimates for the maximal Jacobian norm on compact sets. We also establish an averaged Jacobian-energy bound under one-sided derivative assumptions. Finally, for a kernel-type interacting model satisfying a pairwise dissipativity condition, we prove exponential decay of the expected pairwise variance, yielding mean-square synchronization around the random center of mass.

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1. INTRODUCTION

We consider stochastic differential equations with interaction that were proposed by Andrey A. Dorogovtsev in [Dorogovtsev, 2003],

$$(1) \quad \begin{cases} dX_t(u) &= a(X_t(u), \mu_t) dt + \sum_{\ell=1}^r \sigma_\ell(X_t(u)) dB_t^\ell, \\ X_0(u) &= u, \\ \mu_t &= \mu_0 \circ X_t^{-1}. \end{cases}$$

Here μ_0 is a deterministic initial probability measure, $a : \mathbb{R}^d \times \mathcal{P}_2(\mathbb{R}^d) \rightarrow \mathbb{R}^d$ is the drift, where $\mathcal{P}_2(\mathbb{R}^d)$ denotes the space of Borel probability measures on $\mathbb{R}^d$ with finite second moment. and the diffusion coefficient $\sigma_\ell : \mathbb{R}^d \rightarrow \mathbb{R}^d$, $\ell = 1, \dots, r$, is state-dependent

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and non-constant. Let $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \geq 0}, \mathbb{P})$ be a filtered probability space satisfying the usual conditions and carrying a r -dimensional Brownian motion $B = (B^1, \dots, B^r)$.

Unlike a classical stochastic differential equation, the drift depends on the pushforward of the entire system through the random measure μ_t . This dependence represents the interaction between particles. For each realization $\omega \in \Omega$, the map

$$T_t(\omega, \cdot) : u \mapsto X_t(u, \omega)$$

acts on the initial measure by pushforward,

$$\mu_t(\omega) = \mu_0 \circ T_t(\omega)^{-1},$$

so the evolution is best viewed as a random transport of measures by a stochastic flow.

We establish three types of estimates for measure-valued solutions generated by the stochastic differential equations with interactions. First, we prove pathwise transfer results for logarithmic Sobolev, Poincaré, concentration, and T_2 inequalities under the random transport T_t . Second, under spatial regularity assumptions, we bound moments of the maximal Jacobian norm on compact sets and derive an averaged Jacobian-energy estimate under weaker one-sided conditions. Third, for a kernel-type interacting equation satisfying pairwise dissipativity, we prove exponential decay of the expected pairwise variance and hence mean-square synchronization.

In Section 2 we investigate the Jacobian norm and how it affects the functional inequalities. We write $|\cdot|$ for the Euclidean norm on $\mathbb{R}^d$. For a matrix $M \in \mathbb{R}^{d \times d}$ we set

$$\|M\|_{\text{HS}} := \left(\sum_{i,j=1}^d |M_{ij}|^2 \right)^{1/2}.$$

Fix $\mu_0 \in \mathcal{P}_2(\mathbb{R}^d)$.

Assumption 1 (Well-posedness). *There exists $L < \infty$ such that, for all $x, y \in \mathbb{R}^d$ and $\mu, \nu \in \mathcal{P}_2(\mathbb{R}^d)$,*

$$|a(x, \mu) - a(y, \nu)| + \left(\sum_{\ell=1}^r |\sigma_\ell(x) - \sigma_\ell(y)|^2 \right)^{1/2} \leq L(|x - y| + W_2(\mu, \nu))$$

Under Assumption 1, (1) admits a unique global solution; see [Dorogovtsev, 2023, Theorem 2.3.1]. This assumption remains in force throughout Sections 2 and 3. Additional spatial regularity conditions are stated locally when they are needed. Section 4 concerns a different kernel model and is treated under separate assumptions.

In Section 3, we replace the pathwise supremum over a compact set by an averaged Jacobian energy and derive a corresponding moment estimate.

In Section 4, we consider a specific form of the SDEWI in which the dependence on μ_t remains explicit through the kernel K and the parameter Θ_t .

2. PATHWISE JACOBIAN BOUNDS AND FUNCTIONAL INEQUALITY TRANSFER

This section has two purposes. First, we quantify the spatial distortion of the stochastic flow T_t through its Jacobian J_t . Second, we use this distortion to transfer functional inequalities from the initial measure μ_0 to the random pushforward measure μ_t . The technical proof of the compact-set Jacobian estimate is postponed to the final subsection.

Assumption 2 (Spatial differentiability). *For every $\mu \in \mathcal{P}_2(\mathbb{R}^d)$, the map $x \mapsto a(x, \mu)$ belongs to $C^1(\mathbb{R}^d; \mathbb{R}^d)$, and*

$$\sigma_\ell \in C^1(\mathbb{R}^d; \mathbb{R}^d), \quad \ell = 1, \dots, r.$$

By Assumptions 1 and 2,

$$(2) \quad \|D_x a(x, \mu)\|_{\text{op}} \leq L_x, \quad \sum_{\ell=1}^r |D\sigma_\ell(x)z|^2 \leq L_\sigma^2 |z|^2$$

for all $x, z \in \mathbb{R}^d$ and $\mu \in \mathcal{P}_2(\mathbb{R}^d)$

2.1. The Jacobian flow and distortion constants. We first identify the Jacobian equation for the stochastic flow and introduce the distortion quantities used throughout the section.

For a fixed ω and t , define the flow map $T_t(\omega, u) := X_t(u, \omega)$ and its Jacobian matrix

$$J_t(u) := \partial_u T_t(u) = \partial_u X_t(u) \in \mathbb{R}^{d \times d}.$$

We use the Jacobian SDE from [Dorogovtsev and Weiß, 2023]. For equation (1), it has the following form.

Theorem 2.1 (Jacobian equation). *Assume Assumptions 1 and 2. Then the solution admits a modification such that, for almost every ω and every $t \geq 0$, the map*

$$u \mapsto T_t(\omega, u)$$

is of class C^1 . Its Jacobian

$$J_t(u) := D_u T_t(u)$$

satisfies

$$(3) \quad dJ_t(u) = A_t(u)J_t(u)dt + \sum_{\ell=1}^r C_{\ell,t}(u)J_t(u)dB_t^\ell, \quad J_0(u) = I,$$

where

$$A_t(u) := D_x a(X_t(u), \mu_t), \quad C_{\ell,t}(u) := D\sigma_\ell(X_t(u)).$$

In particular,

$$\|A_t(u)\|_{\text{op}} \leq L_x, \quad \sum_{\ell=1}^r |C_{\ell,t}(u)z|^2 \leq L_\sigma^2 |z|^2$$

for all t, u, z .

Definition 2.1 (Random Jacobian constant). For a nonempty set $K \subset \mathbb{R}^d$, define

$$\mathcal{K}_t(\omega; K) := \sup_{u \in K} \|J_t(u, \omega)\|_{\text{op}}.$$

If K is compact and $u \mapsto J_t(u, \omega)$ is continuous, then $\mathcal{K}_t(\omega; K) < \infty$.

Why $\mathcal{K}_t(\omega; K)$ is finite and relation to global blow-up examples? The counterexamples of Mohammed–Scheutzow show that even for globally Lipschitz coefficients one may fail to have a finite global constant $\sup_{u \in \mathbb{R}^d} \|J_t(u, \omega)\|_{\text{op}}$ for a fixed (t, ω) . In contrast, the quantity $\mathcal{K}_t(\omega; K)$ is taken over a compact set K ; under Assumptions 1 and 2(iii), for each fixed (t, ω) the map $u \mapsto J_t(u, \omega)$ is continuous, hence the supremum over K is finite.

Lemma 2.1 (Flow Lipschitz constant via Jacobian matrix). *For each fixed (t, ω) such that the flow map $u \mapsto T_t(\omega, u) := X_t(u, \omega)$ is of class C^1 ,*

$$\text{Lip}(T_t(\omega)) = \sup_{u \in \mathbb{R}^d} \|J_t(u, \omega)\|_{\text{op}} \in [0, \infty].$$

Proof. Since $u \mapsto T_t(\omega, u)$ is C^1 , the mean value theorem implies

$$|T_t(u) - T_t(v)| \leq \sup_{\theta \in [0, 1]} \|J_t(v + \theta(u - v))\|_{\text{op}} |u - v|.$$

Taking the supremum over $u \neq v$ yields $\text{Lip}(T_t) \leq \sup_u \|J_t(u)\|_{\text{op}}$. Conversely, for any u and unit vector ξ , letting $v = u + \varepsilon \xi$ and sending $\varepsilon \downarrow 0$ gives

$$\|J_t(u)\|_{\text{op}} = \sup_{|\xi|=1} \lim_{\varepsilon \downarrow 0} \frac{|T_t(u + \varepsilon \xi) - T_t(u)|}{\varepsilon} \leq \text{Lip}(T_t),$$

hence equality. $\square$

Pointwise estimates for $J_t(u)$ follow from the boundedness of the coefficients in the linear equation (3). Controlling the compact-set maximal Jacobian norm $\mathcal{K}_t(K)$, however, requires estimates for the spatial increments $J_t(u) - J_t(v)$. The additional regularity needed for this purpose is therefore imposed only in the following theorem.

Theorem 2.2 (Moments of the compact-set maximal Jacobian norm). *Assume Assumptions 1 and 2. Suppose in addition that there exist constants $H_a, H_\sigma < \infty$ such that, for all $x, y, z \in \mathbb{R}^d$ and $\mu \in \mathcal{P}_2(\mathbb{R}^d)$,*

$$(4) \quad |(D_x a(x, \mu) - D_x a(y, \mu))z| \leq H_a |x - y| |z|,$$

$$(5) \quad \sum_{\ell=1}^r |(D\sigma_\ell(x) - D\sigma_\ell(y))z|^2 \leq H_\sigma^2 |x - y|^2 |z|^2.$$

Then, for every compact set $K \subset \mathbb{R}^d$, every $p \geq 1$, and every $t \geq 0$,

$$(6) \quad \mathbb{E}[\mathcal{K}_t(\cdot; K)^p] < \infty.$$

Moreover, there exist constants $C_{p,K}, c_p < \infty$, depending only on $p, K, d, r, L, H_a, H_\sigma$, such that

$$(7) \quad \mathbb{E}[\mathcal{K}_t(\cdot; K)^p] \leq C_{p,K} e^{c_p t}, \quad t \geq 0.$$

The proof is given in Section 2.3.

2.2. Transfer of functional inequalities. In this subsection we record several consequences of the identity

$$T_t(\omega, u) := X_t(u, \omega),$$

which expresses the random measure μ_t as the pushforward of the initial law μ_0 under the stochastic flow.

Definition 2.2 (Logarithmic Sobolev inequality). Denoting by $\mathcal{C}_c^1(\mathbb{R}^d)$ the set of continuously differentiable compactly supported functions from $\mathbb{R}^d$ to $\mathbb{R}$, for $g \geq 0$ with $\int g d\mu = 1$ we define the entropy

$$\text{Ent}_\mu(g) := \int g \log g d\mu.$$

A probability measure μ on $\mathbb{R}^d$ is said to satisfy a logarithmic Sobolev inequality with constant $C > 0$ if for every $h \in \mathcal{C}_c^1(\mathbb{R}^d)$ with $\int h^2 d\mu = 1$ one has

$$(8) \quad \text{Ent}_\mu(h^2) \leq C \int_{\mathbb{R}^d} |\nabla h|^2 d\mu.$$

Corollary 2.1 (Pathwise LSI with the Jacobian matrix constant). *If μ_0 satisfies $\text{LSI}(C_0)$ and $\text{supp}(\mu_0) \subset K$ for some compact K , then for every $t \geq 0$ and a.s. ω ,*

$$\mu_t(\omega) = \mu_0 \circ T_t(\omega)^{-1} \implies \mu_t(\omega) \text{ satisfies } \text{LSI}(C_0 \mathcal{K}_t(\omega; K)^2).$$

We next record the basic concentration-transfer principle under a random Lipschitz map.

Proposition 2.1 (Pathwise concentration transfer). *Let $K := \text{supp}(\mu_0)$ and assume that for a.s. ω the map $T_t(\omega, \cdot)$ is Lipschitz on K , with Lipschitz constant*

$$L_t(\omega; K) := \sup_{\substack{u, v \in K \\ u \neq v}} \frac{|T_t(\omega, u) - T_t(\omega, v)|}{|u - v|}.$$

Assume that μ_0 satisfies the concentration inequality

$$(9) \quad \mu_0(f - \mu_0(f) \geq r) \leq \Psi(r), \quad r \geq 0,$$

for every 1-Lipschitz function $f : \mathbb{R}^d \rightarrow \mathbb{R}$, where $\Psi : [0, \infty) \rightarrow [0, 1]$ is nonincreasing. Then for a.s. ω ,

$$(10) \quad \mu_t(\omega)(g - \mu_t(\omega)(g) \geq r) \leq \Psi\left(\frac{r}{L_t(\omega; K)}\right), \quad r \geq 0,$$

for every 1-Lipschitz function $g : \mathbb{R}^d \rightarrow \mathbb{R}$.

Proof. Fix ω such that $T_t(\omega, \cdot)$ is Lipschitz on K , and write $L := L_t(\omega; K) \in [0, \infty)$. Let $g : \mathbb{R}^d \rightarrow \mathbb{R}$ be 1-Lipschitz.

If $L = 0$, then $T_t(\omega, \cdot)$ is constant on K , hence $\mu_t(\omega) = \mu_0 \circ T_t(\omega)^{-1}$ is a Dirac mass and $\mu_t(\omega)(g - \mu_t(\omega)(g) \geq r) = 0$ for every $r > 0$. Thus equation (10) is immediate.

Assume now $L > 0$ and define on K

$$h(u) := \frac{g(T_t(\omega, u))}{L}.$$

Then h is 1-Lipschitz on K . By the McShane extension theorem, h extends to a 1-Lipschitz function $\tilde{h} : \mathbb{R}^d \rightarrow \mathbb{R}$. Applying equation (9) to $\tilde{h}$ (and using that μ_0 is supported on K) gives

$$\mu_0(h - \mu_0(h) \geq s) \leq \Psi(s), \quad s \geq 0.$$

Since $\mu_t(\omega) = \mu_0 \circ T_t(\omega)^{-1}$,

$$\mu_t(\omega)(g) = \mu_0(g \circ T_t(\omega, \cdot)) = L\mu_0(h).$$

Therefore, for $r \geq 0$,

$$\begin{aligned} \mu_t(\omega)(g - \mu_t(\omega)(g) \geq r) &= \mu_0(g \circ T_t(\omega, \cdot) - \mu_0(g \circ T_t(\omega, \cdot)) \geq r) \\ &= \mu_0(h - \mu_0(h) \geq r/L) \\ &\leq \Psi\left(\frac{r}{L}\right) = \Psi\left(\frac{r}{L_t(\omega; K)}\right). \end{aligned}$$

This proves equation (10). $\square$

Remark 2.1 (Concentration as a corollary of LSI). If μ_0 satisfies a logarithmic Sobolev inequality, then it satisfies Gaussian concentration. Hence Corollary 2.1 yields, in particular, a pathwise concentration inequality for $\mu_t(\omega)$ with a random constant controlled by $\mathcal{K}_t(\omega; K)^2$.

Definition 2.3 (Poincaré inequality). A probability measure μ on $\mathbb{R}^d$ is said to satisfy a Poincaré inequality with constant $C_P > 0$, denoted by $\text{PI}(C_P)$, if for every $h \in \mathcal{C}_c^1(\mathbb{R}^d)$,

$$(11) \quad \text{Var}_\mu(h) := \int_{\mathbb{R}^d} h^2 d\mu - \left(\int_{\mathbb{R}^d} h d\mu \right)^2 \leq C_P \int_{\mathbb{R}^d} |\nabla h|^2 d\mu.$$

We first record the stability of the Poincaré inequality under Lipschitz pushforward.

Proposition 2.2 (Poincaré inequality under a Lipschitz pushforward). *Let μ be a probability measure on $\mathbb{R}^d$ satisfying $\text{PI}(C_P)$, and let $T : \mathbb{R}^d \rightarrow \mathbb{R}^d$ be a Lipschitz map with Lipschitz constant $\text{Lip}(T)$. Then the pushforward measure $\nu := \mu \circ T^{-1}$ satisfies*

$$\text{PI}(C_P \text{Lip}(T)^2).$$

Proof. Let $g \in C_c^1(\mathbb{R}^d)$ and set $f := g \circ T$. Then

$$\int_{\mathbb{R}^d} g \, d\nu = \int_{\mathbb{R}^d} g(T(u)) \, \mu(du) = \int_{\mathbb{R}^d} f(u) \, \mu(du),$$

and similarly

$$\int_{\mathbb{R}^d} g^2 \, d\nu = \int_{\mathbb{R}^d} f^2 \, d\mu.$$

Hence

$$\text{Var}_\nu(g) = \text{Var}_\mu(f).$$

Applying the Poincaré inequality for μ gives

$$\text{Var}_\nu(g) = \text{Var}_\mu(f) \leq C_P \int_{\mathbb{R}^d} |\nabla f(u)|^2 \, \mu(du).$$

Since $f = g \circ T$ and T is Lipschitz, we have a.e.

$$|\nabla f(u)| \leq \text{Lip}(T) |\nabla g(T(u))|.$$

Therefore

$$\text{Var}_\nu(g) \leq C_P \text{Lip}(T)^2 \int_{\mathbb{R}^d} |\nabla g(T(u))|^2 \, \mu(du) = C_P \text{Lip}(T)^2 \int_{\mathbb{R}^d} |\nabla g(x)|^2 \, \nu(dx).$$

This proves the claim. $\square$

Corollary 2.2 (Pathwise Poincaré inequality). *If μ_0 satisfies $\text{PI}(C_P)$ and $\text{supp}(\mu_0) \subset K$ for some compact set $K \subset \mathbb{R}^d$, then for every $t \geq 0$ and a.s. ω ,*

$$\mu_t(\omega) = \mu_0 \circ T_t(\omega)^{-1} \implies \mu_t(\omega) \text{ satisfies } \text{PI}(C_P \mathcal{K}_t(\omega; K)^2).$$

Proof. By Proposition 2.2, it suffices to bound the Lipschitz constant of $T_t(\omega, \cdot)$ on K . Since K is compact and $u \mapsto T_t(\omega, u)$ is of class C^1 , the mean value theorem gives

$$|T_t(\omega, u) - T_t(\omega, v)| \leq \sup_{z \in K} \|J_t(z, \omega)\|_{\text{op}} |u - v| = \mathcal{K}_t(\omega; K) |u - v|, \quad u, v \in K.$$

Hence $\text{Lip}(T_t(\omega)|_K) \leq \mathcal{K}_t(\omega; K)$, and the result follows. $\square$

Definition 2.4 (Talagrand transport inequality). Let $p \geq 1$. A probability measure μ on $\mathbb{R}^d$ is said to satisfy the Talagrand transport inequality $T_p(C_T)$ with constant $C_T > 0$ if for every probability measure ν on $\mathbb{R}^d$ with finite p -th moment,

$$(12) \quad W_p(\nu, \mu)^2 \leq 2C_T \text{Ent}(\nu \mid \mu),$$

where

$$\text{Ent}(\nu \mid \mu) = \begin{cases} \int_{\mathbb{R}^d} \log\left(\frac{d\nu}{d\mu}\right) d\nu, & \nu \ll \mu, \\ +\infty, & \text{otherwise.} \end{cases}$$

We next record the analogous stability property for Talagrand transport inequalities.

Proposition 2.3 (Transport inequality under a Lipschitz pushforward). *Let μ be a probability measure on $\mathbb{R}^d$ satisfying $T_p(C_T)$ for some $p \geq 1$, and let $T : \mathbb{R}^d \rightarrow \mathbb{R}^d$ be a Lipschitz map with Lipschitz constant $\text{Lip}(T)$. Then the pushforward measure $\eta := \mu \circ T^{-1}$ satisfies*

$$T_p(C_T \text{Lip}(T)^2).$$

Proof. Let λ be any probability measure on $\mathbb{R}^d$. If $\lambda \not\ll \eta$, then $\text{Ent}(\lambda \mid \eta) = +\infty$ and the inequality is trivial. Assume therefore that $\lambda \ll \eta$.

Choose a probability measure $\tilde{\lambda}$ on $\mathbb{R}^d$ such that

$$\tilde{\lambda} \circ T^{-1} = \lambda \quad \text{and} \quad \text{Ent}(\tilde{\lambda} \mid \mu) = \text{Ent}(\lambda \mid \eta).$$

For instance, one may take the measure

$$\tilde{\lambda}(du) := \frac{d\lambda}{d\eta}(T(u)) \, \mu(du).$$

Indeed,

$$(\tilde{\lambda} \circ T^{-1})(A) = \int_{\mathbb{R}^d} \mathbf{1}_{\{T(u) \in A\}} \frac{d\lambda}{d\eta}(T(u)) \mu(du) = \int_A \frac{d\lambda}{d\eta}(x) \eta(dx) = \lambda(A),$$

and

$$\text{Ent}(\tilde{\lambda} \mid \mu) = \int_{\mathbb{R}^d} \log \left(\frac{d\lambda}{d\eta}(T(u)) \right) \tilde{\lambda}(du) = \int_{\mathbb{R}^d} \log \left(\frac{d\lambda}{d\eta}(x) \right) \lambda(dx) = \text{Ent}(\lambda \mid \eta).$$

Now let π be a coupling of $\tilde{\lambda}$ and μ . Then $\pi \circ (T, T)^{-1}$ is a coupling of λ and η , so by Lipschitz continuity of T ,

$$\begin{aligned} & \int_{\mathbb{R}^d \times \mathbb{R}^d} |x - y|^p d(\pi \circ (T, T)^{-1})(x, y) \\ &= \int_{\mathbb{R}^d \times \mathbb{R}^d} |T(u) - T(v)|^p d\pi(u, v) \\ &\leq \text{Lip}(T)^p \int |u - v|^p d\pi(u, v). \end{aligned}$$

Taking the infimum over all couplings π of $\tilde{\lambda}$ and μ yields

$$W_p(\lambda, \eta) \leq \text{Lip}(T) W_p(\tilde{\lambda}, \mu).$$

Since μ satisfies $T_p(C_T)$,

$$W_p(\tilde{\lambda}, \mu)^2 \leq 2C_T \text{Ent}(\tilde{\lambda} \mid \mu) = 2C_T \text{Ent}(\lambda \mid \eta).$$

Combining the two inequalities, we obtain

$$W_p(\lambda, \eta)^2 \leq \text{Lip}(T)^2 W_p(\tilde{\lambda}, \mu)^2 \leq 2C_T \text{Lip}(T)^2 \text{Ent}(\lambda \mid \eta),$$

which proves that η satisfies $T_p(C_T \text{Lip}(T)^2)$. $\square$

2.3. Proof of the compact-set Jacobian estimate. We begin with a standard moment estimate for linear stochastic equations.

Lemma 2.2 (Moment estimate for linear stochastic equations). *Let Y_t solve*

$$dY_t = A_t Y_t dt + \sum_{\ell=1}^r C_{\ell,t} Y_t dB_t^\ell, \quad Y_0 = y.$$

Assume that, for every $z \in \mathbb{R}^d$,

$$\langle z, A_t z \rangle \leq L_x |z|^2, \quad \sum_{\ell=1}^r |C_{\ell,t} z|^2 \leq L_\sigma^2 |z|^2.$$

Then, for every $q \geq 1$,

$$\mathbb{E}|Y_t|^{2q} \leq |y|^{2q} \exp \{ q(2L_x + (2q-1)L_\sigma^2)t \}.$$

Consequently,

$$\mathbb{E}\|J_t(u)\|_{\text{op}}^{2q} \leq d^q \exp \{ q(2L_x + (2q-1)L_\sigma^2)t \}.$$

Proof. Apply Itô's formula to $|Y_t|^2$, use the bounds on A_t and $C_t^{(k)}$, and then apply the exponential martingale estimate. The matrix estimate follows by applying the vector estimate to the columns of $J_t(u)$ and using $\|J_t(u)\|_{\text{op}} \leq \|J_t(u)\|_{\text{HS}}$. $\square$

Although the drift depends on the random measure μ_t , differentiation with respect to the initial label acts only on the spatial argument $X_t(u)$. Because of that fact we can evaluate $\|J_t(u)\|$ using Kunita's theorem for the stochastic flows.

Under Assumptions 1 and 2, standard results on stochastic flows (see, e.g., [Kunita, 1986, Theorem 1.4.1]) yield a modification such that for a.s. ω and each $t \geq 0$ the map $u \mapsto T_t(\omega, u)$ is $\mathcal{C}^1$.

Proof of Theorem 2.2.

Step 1: Pointwise moments and flow increments. Fix a compact set $K \subset \mathbb{R}^d$, $t \geq 0$, and $p \geq 1$. Choose an even integer $m > \max\{2p, 2d\}$. It is enough to prove (7) with p replaced by m , because then Hölder's inequality gives

$$\mathbb{E} \mathcal{K}_t(\cdot; K)^p \leq (\mathbb{E} \mathcal{K}_t(\cdot; K)^m)^{p/m}.$$

Fix $u, v \in \mathbb{R}^d$ and set

$$\Delta X_t := X_t(u) - X_t(v).$$

Since the same random measure μ_t appears in both equations, subtraction yields

$$d\Delta X_t = (a(X_t(u), \mu_t) - a(X_t(v), \mu_t)) dt + (\sigma(X_t(u)) - \sigma(X_t(v))) dB_t.$$

For $u \neq v$, define matrix-valued processes

$$\tilde{A}_t(u, v) := \int_0^1 \partial_x a(X_t(v) + \theta(X_t(u) - X_t(v)), \mu_t) d\theta$$

and, for $k = 1, \dots, d$,

$$\tilde{C}_t^{(k)}(u, v) := \int_0^1 \partial_x \sigma^{(k)}(X_t(v) + \theta(X_t(u) - X_t(v))) d\theta,$$

where $\sigma^{(k)}$ denotes the k -th column of σ . By the fundamental theorem of calculus,

$$a(X_t(u), \mu_t) - a(X_t(v), \mu_t) = \tilde{A}_t(u, v) \Delta X_t,$$

and

$$\sigma^{(k)}(X_t(u)) - \sigma^{(k)}(X_t(v)) = \tilde{C}_t^{(k)}(u, v) \Delta X_t.$$

Hence ΔX_t satisfies the linear SDE

$$d\Delta X_t = \tilde{A}_t(u, v) \Delta X_t dt + \sum_{k=1}^d \tilde{C}_t^{(k)}(u, v) \Delta X_t dB_t^k, \quad \Delta X_0 = u - v.$$

Moreover, by Assumption ??,

$$\|\tilde{A}_t(u, v)\|_{\text{op}} \leq L, \quad \|\tilde{C}_t^{(k)}(u, v)\|_{\text{op}} \leq L_\sigma.$$

If $u = v$, then $\Delta X_t \equiv 0$, so the desired estimate is trivial. Assume therefore that $u \neq v$, and set

$$Z_t := \frac{\Delta X_t}{|u - v|}, \quad \xi := \frac{u - v}{|u - v|}.$$

Then $|\xi| = 1$, $Z_0 = \xi$, and Z_t solves

$$dZ_t = \tilde{A}_t(u, v) Z_t dt + \sum_{k=1}^d \tilde{C}_t^{(k)}(u, v) Z_t dB_t^k.$$

Applying Corollary 2.2 to this linear equation with exponent $q = m/2$, we obtain

$$\mathbb{E}|Z_t|^m \leq C_m e^{c_m t},$$

for constants $C_m, c_m < \infty$ depending only on m, d, L, L_σ . Multiplying by $|u - v|^m$ yields

$$(13) \quad \mathbb{E}|X_t(u) - X_t(v)|^m \leq C_m e^{c_m t} |u - v|^m, \quad u, v \in \mathbb{R}^d.$$

By Corollary 2.2, for every $q \geq 1$ and every $u \in \mathbb{R}^d$,

$$\mathbb{E} \|J_t(u)\|_{\text{op}}^{2q} \leq d^q \exp \left(q(2L + dL_\sigma^2)t + 2q^2 dL_\sigma^2 t \right).$$

In particular, taking $q = m/2$ and $q = m$, we obtain

$$(14) \quad \mathbb{E} \|J_t(u)\|_{\text{op}}^m \leq C_m e^{c_m t}, \quad \mathbb{E} \|J_t(u)\|_{\text{op}}^{2m} \leq C_m e^{c_m t},$$

uniformly in $u \in \mathbb{R}^d$.

Step 2: Spatial increments of the Jacobian. Fix $u, v \in \mathbb{R}^d$ and define

$$D_t := J_t(u) - J_t(v).$$

Subtracting the Jacobian equations (3) at u and v , we obtain
(15)

$$\begin{aligned} dD_t &= A_t(u)D_t dt + \sum_{k=1}^d C_t^{(k)}(u)D_t dB_t^k \\ &\quad + (A_t(u) - A_t(v))J_t(v) dt + \sum_{k=1}^d (C_t^{(k)}(u) - C_t^{(k)}(v))J_t(v) dB_t^k, \quad D_0 = 0, \end{aligned}$$

where

$$A_t(w) := \partial_x a(X_t(w), \mu_t), \quad C_t^{(k)}(w) := \partial_x \sigma^{(k)}(X_t(w)).$$

By Assumptions ?? and ??,

$$\|A_t(w)\|_{\text{op}} \leq L, \quad \|C_t^{(k)}(w)\|_{\text{op}} \leq L_\sigma,$$

and

$$(16) \quad \|A_t(u) - A_t(v)\|_{\text{op}} \leq L^{(2)}|X_t(u) - X_t(v)|, \quad \|C_t^{(k)}(u) - C_t^{(k)}(v)\|_{\text{op}} \leq L_\sigma^{(2)}|X_t(u) - X_t(v)|.$$

We estimate

$$\Phi(t) := \mathbb{E} \sup_{s \leq t} \|D_s\|_{\text{HS}}^m.$$

Set

$$F_s := (A_s(u) - A_s(v))J_s(v), \quad G_s^{(k)} := (C_s^{(k)}(u) - C_s^{(k)}(v))J_s(v).$$

Then

$$dD_t = A_t(u)D_t dt + \sum_{k=1}^d C_t^{(k)}(u)D_t dB_t^k + F_t dt + \sum_{k=1}^d G_t^{(k)} dB_t^k, \quad D_0 = 0.$$

Hence, for $s \leq t$,

$$D_s = \int_0^s A_r(u)D_r dr + \sum_{k=1}^d \int_0^s C_r^{(k)}(u)D_r dB_r^k + \int_0^s F_r dr + \sum_{k=1}^d \int_0^s G_r^{(k)} dB_r^k.$$

Using $(a_1 + \dots + a_4)^m \leq C_m(a_1^m + \dots + a_4^m)$, we obtain

$$\begin{aligned} \sup_{s \leq t} \|D_s\|_{\text{HS}}^m &\leq C_m \left[\sup_{s \leq t} \left\| \int_0^s A_r(u)D_r dr \right\|_{\text{HS}}^m + \sup_{s \leq t} \left\| \sum_{k=1}^d \int_0^s C_r^{(k)}(u)D_r dB_r^k \right\|_{\text{HS}}^m \right. \\ &\quad \left. + \sup_{s \leq t} \left\| \int_0^s F_r dr \right\|_{\text{HS}}^m + \sup_{s \leq t} \left\| \sum_{k=1}^d \int_0^s G_r^{(k)} dB_r^k \right\|_{\text{HS}}^m \right]. \end{aligned}$$

Taking expectations, we estimate the four terms separately.

For the drift term involving $A_r(u)D_r$,

$$\sup_{s \leq t} \left\| \int_0^s A_r(u)D_r dr \right\|_{\text{HS}} \leq L \int_0^t \|D_r\|_{\text{HS}} dr,$$

hence by Jensen's inequality,

$$\mathbb{E} \sup_{s \leq t} \left\| \int_0^s A_r(u)D_r dr \right\|_{\text{HS}}^m \leq L^m t^{m-1} \int_0^t \mathbb{E} \sup_{\rho \leq r} \|D_\rho\|_{\text{HS}}^m dr.$$

For the stochastic term involving $C_r^{(k)}(u)D_r$, the BDG inequality yields

$$\begin{aligned} \mathbb{E} \sup_{s \leq t} \left\| \sum_{k=1}^d \int_0^s C_r^{(k)}(u) D_r dB_r^k \right\|_{\text{HS}}^m &\leq C_m \mathbb{E} \left(\int_0^t \sum_{k=1}^d \|C_r^{(k)}(u) D_r\|_{\text{HS}}^2 dr \right)^{m/2} \\ &\leq C_m \mathbb{E} \left(\int_0^t dL_\sigma^2 \|D_r\|_{\text{HS}}^2 dr \right)^{m/2}. \end{aligned}$$

Since $m \geq 2$, Jensen's inequality gives

$$\left(\int_0^t \|D_r\|_{\text{HS}}^2 dr \right)^{m/2} \leq t^{\frac{m}{2}-1} \int_0^t \sup_{\rho \leq r} \|D_\rho\|_{\text{HS}}^m dr,$$

and therefore

$$\mathbb{E} \sup_{s \leq t} \left\| \sum_{k=1}^d \int_0^s C_r^{(k)}(u) D_r dB_r^k \right\|_{\text{HS}}^m \leq C_m t^{\frac{m}{2}-1} \int_0^t \Phi(r) dr.$$

Next,

$$\sup_{s \leq t} \left\| \int_0^s F_r dr \right\|_{\text{HS}}^m \leq t^{m-1} \int_0^t \|F_r\|_{\text{HS}}^m dr,$$

hence

$$\mathbb{E} \sup_{s \leq t} \left\| \int_0^s F_r dr \right\|_{\text{HS}}^m \leq t^{m-1} \int_0^t \mathbb{E} \|F_r\|_{\text{HS}}^m dr.$$

Finally, applying the BDG inequality to the martingale term involving $G_r^{(k)}$, we obtain

$$\begin{aligned} \mathbb{E} \sup_{s \leq t} \left\| \sum_{k=1}^d \int_0^s G_r^{(k)} dB_r^k \right\|_{\text{HS}}^m &\leq C_m \mathbb{E} \left(\int_0^t \sum_{k=1}^d \|G_r^{(k)}\|_{\text{HS}}^2 dr \right)^{m/2} \\ &\leq C_m t^{\frac{m}{2}-1} \int_0^t \sum_{k=1}^d \mathbb{E} \|G_r^{(k)}\|_{\text{HS}}^m dr. \end{aligned}$$

Combining the four estimates, we obtain

$$\begin{aligned} \Phi(t) &\leq C_m (1 + t^{m-1} + t^{\frac{m}{2}-1}) \int_0^t \Phi(r) dr \\ (17) \quad &+ C_m (1 + t^{m-1} + t^{\frac{m}{2}-1}) \int_0^t \left(\mathbb{E} \|F_r\|_{\text{HS}}^m + \sum_{k=1}^d \mathbb{E} \|G_r^{(k)}\|_{\text{HS}}^m \right) dr. \end{aligned}$$

We now estimate the forcing terms. By (16),

$$\|F_r\|_{\text{HS}} \leq L^{(2)} |X_r(u) - X_r(v)| \|J_r(v)\|_{\text{HS}},$$

and

$$\|G_r^{(k)}\|_{\text{HS}} \leq L_\sigma^{(2)} |X_r(u) - X_r(v)| \|J_r(v)\|_{\text{HS}}.$$

Hence, by Hölder's inequality,

$$\begin{aligned} \mathbb{E} \|F_r\|_{\text{HS}}^m &\leq C \left(\mathbb{E} |X_r(u) - X_r(v)|^{2m} \right)^{1/2} \left(\mathbb{E} \|J_r(v)\|_{\text{HS}}^{2m} \right)^{1/2}, \\ \mathbb{E} \|G_r^{(k)}\|_{\text{HS}}^m &\leq C \left(\mathbb{E} |X_r(u) - X_r(v)|^{2m} \right)^{1/2} \left(\mathbb{E} \|J_r(v)\|_{\text{HS}}^{2m} \right)^{1/2}. \end{aligned}$$

Since $\|M\|_{\text{HS}} \leq \sqrt{d} \|M\|_{\text{op}}$, (13) and (14) yield

$$\mathbb{E} \|F_r\|_{\text{HS}}^m + \sum_{k=1}^d \mathbb{E} \|G_r^{(k)}\|_{\text{HS}}^m \leq C_m e^{c_m r} |u - v|^m.$$

Substituting this into (17), we obtain

$$\Phi(t) \leq C_m (1 + t^{m-1} + t^{\frac{m}{2}-1}) \int_0^t \Phi(r) dr + C_m (1 + t^{m-1} + t^{\frac{m}{2}-1}) e^{c_m t} |u - v|^m.$$

By Gronwall's lemma,

$$\Phi(t) \leq C_m(1 + t^{m-1} + t^{\frac{m}{2}-1})e^{c_m t}|u - v|^m.$$

Absorbing the polynomial factor into the exponential by enlarging c_m , we get

$$\Phi(t) \leq C_m e^{c_m t}|u - v|^m.$$

In particular,

$$\mathbb{E} \|D_t\|_{\text{HS}}^m \leq \mathbb{E} \sup_{s \leq t} \|D_s\|_{\text{HS}}^m \leq C_m e^{c_m t}|u - v|^m.$$

Since $\|D_t\|_{\text{op}} \leq \|D_t\|_{\text{HS}}$, we conclude that

$$(18) \quad \mathbb{E} \|J_t(u) - J_t(v)\|_{\text{op}}^m \leq C_m e^{c_m t}|u - v|^m, \quad u, v \in \mathbb{R}^d.$$

Step 3: Kolmogorov continuity and the compact supremum. Fix $\beta \in (0, 1 - d/m)$. Since $m > d$, the increment estimate (18) implies, by the Kolmogorov continuity theorem applied in the spatial variable u , that for each fixed $t \geq 0$ there exists a modification of $u \mapsto J_t(u)$ whose sample paths on K are β -Hölder continuous. Moreover, by a quantitative version of the Kolmogorov theorem, one may choose such a modification, still denoted by J_t , so that

$$(19) \quad \mathbb{E} [J_t]_{\beta; K}^m \leq C_{m, \beta, K} e^{c_m t},$$

where

$$[J_t]_{\beta; K} := \sup_{\substack{u, v \in K \\ u \neq v}} \frac{\|J_t(u) - J_t(v)\|_{\text{op}}}{|u - v|^\beta}.$$

Choose any point $u_0 \in K$. Then for every $u \in K$,

$$\|J_t(u)\|_{\text{op}} \leq \|J_t(u_0)\|_{\text{op}} + [J_t]_{\beta; K}|u - u_0|^\beta \leq \|J_t(u_0)\|_{\text{op}} + [J_t]_{\beta; K}(\text{diam } K)^\beta.$$

Therefore,

$$\mathcal{K}_t(\omega; K) = \sup_{u \in K} \|J_t(u, \omega)\|_{\text{op}} \leq \|J_t(u_0, \omega)\|_{\text{op}} + [J_t(\omega)]_{\beta; K}(\text{diam } K)^\beta.$$

Taking m -th moments and using (14) and (19), we obtain

$$\mathbb{E} \mathcal{K}_t(\cdot; K)^m \leq C_{m, K} e^{c_m t}.$$

As explained at the beginning, this implies

$$\mathbb{E} \mathcal{K}_t(\cdot; K)^p \leq C_{p, K} e^{c_p t}, \quad t \geq 0,$$

for every $p \geq 1$. In particular, (6) holds. $\square$

3. THE AVERAGED JACOBIAN

The preceding pathwise estimates use the quantity $\mathcal{K}_t(\omega; K)$ which controls the worst possible stretching of the flow on the compact set K . Consider instead the averaged Jacobian for the transported measure $\mu_t = \mu_0 \circ T_t^{-1}$,

$$\mathcal{E}_t := \int_{\mathbb{R}^d} \|J_t(u)\|_{\text{HS}}^2 \mu_0(du).$$

It measures average deformation of the flow over the initial mass, rather than the maximal deformation on the whole support.

Proposition 3.1 (Averaged Jacobian energy estimate). *Assume that the flow is differentiable in the initial label and that there exist constants $L_{\text{sym}}, \Gamma < \infty$ such that*

$$(20) \quad \langle z, \partial_x a(x, \mu) z \rangle \leq L_{\text{sym}} |z|^2, \quad x, z \in \mathbb{R}^d, \quad \mu \in \mathcal{P}_2(\mathbb{R}^d),$$

and

$$(21) \quad \sum_{k=1}^d |\partial_x \sigma^{(k)}(x) z|^2 \leq \Gamma^2 |z|^2, \quad x, z \in \mathbb{R}^d.$$

Then

$$(22) \quad \mathbb{E} \mathcal{E}_t \leq d \exp\{(2L_{\text{sym}} + \Gamma^2)t\}.$$

Proof. Recall that

$$dJ_t(u) = A_t(u) J_t(u) dt + \sum_{k=1}^d C_t^{(k)}(u) J_t(u) dB_t^k,$$

where

$$A_t(u) = \partial_x a(X_t(u), \mu_t), \quad C_t^{(k)}(u) = \partial_x \sigma^{(k)}(X_t(u)).$$

Applying Itô's formula to $\|J_t(u)\|_{\text{HS}}^2$ gives

$$\begin{aligned} d\|J_t(u)\|_{\text{HS}}^2 &= \left[2\langle J_t(u), A_t(u) J_t(u) \rangle_{\text{HS}} + \sum_{k=1}^d \|C_t^{(k)}(u) J_t(u)\|_{\text{HS}}^2 \right] dt \\ &\quad + 2 \sum_{k=1}^d \langle J_t(u), C_t^{(k)}(u) J_t(u) \rangle_{\text{HS}} dB_t^k. \end{aligned}$$

Since

$$\langle J, A_t J \rangle_{\text{HS}} = \sum_{i=1}^d \langle J e_i, A_t J e_i \rangle,$$

condition equation (20) gives

$$2\langle J, A_t J \rangle_{\text{HS}} \leq 2L_{\text{sym}} \|J\|_{\text{HS}}^2.$$

Similarly, equation (21) gives

$$\sum_{k=1}^d \|C_t^{(k)} J\|_{\text{HS}}^2 = \sum_{i=1}^d \sum_{k=1}^d |C_t^{(k)} J e_i|^2 \leq \Gamma^2 \|J\|_{\text{HS}}^2.$$

Integrating with respect to $\mu_0(du)$ and taking expectations removes the martingale term, hence

$$\frac{d}{dt} \mathbb{E} \mathcal{E}_t \leq (2L_{\text{sym}} + \Gamma^2) \mathbb{E} \mathcal{E}_t.$$

Since $J_0(u) = I$, we have $\mathcal{E}_0 = \|I\|_{\text{HS}}^2 = d$. Gronwall's lemma proves equation (22). $\square$

To estimate $\mathbb{E} \sup_{u \in K} \|J_t(u)\|^p$, one must control increments $J_t(u) - J_t(v)$, which requires assumptions on the spatial regularity of $\partial_x a$ and $\partial \sigma$. The averaged quantity $\mathcal{E}_t$ does not compare two labels. It only estimates each $J_t(u)$ and integrates in u , so the one-sided bounds equations (20) and (21) are sufficient.

4. INTERNAL FLUCTUATIONS

This section concerns a separate kernel-type interacting model. In contrast to (1), both the drift and diffusion may depend on the order parameter Θ_t . Consequently, the assumptions of Sections 2 and 3 are not imposed here.

$$(23) \quad \begin{cases} dX_t(u) &= \left[b(X_t(u), \Theta_t) + \int_{\mathbb{R}^d} K(X_t(u), X_t(v), \Theta_t) \mu_0(dv) \right] dt \\ &\quad + \sum_{\ell=1}^m \sigma_\ell(X_t(u), \Theta_t) dB_t^\ell, \\ X_0(u) &= u, \quad u \in \mathbb{R}^d, \\ \mu_t &= \mu_0 \circ X_t^{-1}. \end{cases}$$

where

$$\theta(\mu) := \int_{\mathbb{R}^d} \Phi(x) \mu(dx),$$

Assumption 3 (Well-posedness of Equation (23)). *The maps b, K, σ_ℓ, Φ are globally Lipschitz and have at most linear growth. More explicitly, there exists $L < \infty$ such that*

$$\begin{aligned} |\Phi(x) - \Phi(y)| &\leq L|x - y|, \\ |b(x, \theta) - b(y, \eta)| &\leq L(|x - y| + |\theta - \eta|), \\ |K(x, z, \theta) - K(y, w, \eta)| &\leq L(|x - y| + |z - w| + |\theta - \eta|), \end{aligned}$$

and

$$\sum_{\ell=1}^m |\sigma_\ell(x, \theta) - \sigma_\ell(y, \eta)|^2 \leq L^2(|x - y|^2 + |\theta - \eta|^2).$$

Under Assumption 3, equation (23) admits a unique global continuous solution. This section studies the pairwise energy

$$\mathcal{D}_t := \iint_{\mathbb{R}^d \times \mathbb{R}^d} |X_t(u) - X_t(v)|^2 \mu_0(du) \mu_0(dv).$$

4.1. Dissipativity assumptions. Equivalently, if

$$\bar{X}_t := \int_{\mathbb{R}^d} X_t(u) \mu_0(du),$$

and

$$\mathcal{V}_t := \int_{\mathbb{R}^d} |X_t(u) - \bar{X}_t|^2 \mu_0(du),$$

then

$$\mathcal{D}_t = 2\mathcal{V}_t.$$

Indeed,

$$\begin{aligned} \mathcal{D}_t &= \iint |X_t(u) - X_t(v)|^2 \mu_0(du) \mu_0(dv) \\ &= \iint (|X_t(u)|^2 + |X_t(v)|^2 - 2\langle X_t(u), X_t(v) \rangle) \mu_0(du) \mu_0(dv) \\ &= 2 \int |X_t(u)|^2 \mu_0(du) - 2 \left| \int X_t(u) \mu_0(du) \right|^2 \\ &= 2 \int |X_t(u) - \bar{X}_t|^2 \mu_0(du) = 2\mathcal{V}_t. \end{aligned}$$

We assume the following dissipativity condition.

Assumption 4 (Pairwise dissipativity). *There exists $\lambda > 0$ such that for every $x, y \in \mathbb{R}^d$, every $\theta \in \mathbb{R}^q$, and every $\nu \in \mathcal{P}_2(\mathbb{R}^d)$,*

$$2 \left\langle x - y, b(x, \theta) - b(y, \theta) + \int_{\mathbb{R}^d} [K(x, z, \theta) - K(y, z, \theta)] \nu(dz) \right\rangle + \sum_{\ell=1}^m |\sigma_\ell(x, \theta) - \sigma_\ell(y, \theta)|^2 \leq -\lambda |x - y|^2.$$

The dissipativity condition covers several basic models.

Example 4.1 (Ornstein–Uhlenbeck SDEWI). Consider

$$dX_t(u) = -\lambda_0 (X_t(u) - m_t) dt + \sigma(X_t(u)) dB_t, \quad m_t = \int_{\mathbb{R}^d} X_t(v) \mu_0(dv).$$

If

$$\|\sigma(x) - \sigma(y)\|_{\text{HS}} \leq L_\sigma |x - y|,$$

then

$$\mathbb{E} \mathcal{D}_t \leq e^{-(2\lambda_0 - L_\sigma^2)t} \mathcal{D}_0.$$

Thus the cloud contracts whenever $2\lambda_0 > L_\sigma^2$.

Example 4.2 (Gradient systems). Take

$$b(x, \theta) = -\nabla V(x), \quad K(x, y, \theta) = -\nabla W(x - y).$$

Then

$$dX_t(u) = \left[-\nabla V(X_t(u)) - \int_{\mathbb{R}^d} \nabla W(X_t(u) - X_t(v)) \mu_0(dv) \right] dt + \sigma(X_t(u)) dB_t.$$

If

$$D^2V \geq \rho I, \quad D^2W \geq \kappa I, \quad \|\sigma(x) - \sigma(y)\| \leq L_\sigma |x - y|,$$

then

$$\mathbb{E} \mathcal{D}_t \leq e^{-(2\rho + 2\kappa - L_\sigma^2)t} \mathcal{D}_0.$$

Example 4.3 (Nonlinear statistics). Let

$$\Theta_t = \int_{\mathbb{R}^d} \psi(X_t(v)) \mu_0(dv),$$

and consider

$$dX_t(u) = -\lambda_0 (X_t(u) - \Theta_t) dt + \sigma(X_t(u), \Theta_t) dB_t.$$

For two labels u, v , the order parameter cancels in the drift difference:

$$-\lambda_0 (X_t(u) - \Theta_t) + \lambda_0 (X_t(v) - \Theta_t) = -\lambda_0 (X_t(u) - X_t(v)).$$

4.2. Proof.

Theorem 4.1 (Synchronization under dissipativity assumptions). *Assume that $\mu_0 \in \mathcal{P}_2(\mathbb{R}^d)$ and that Assumption 4 holds. Then*

$$\mathbb{E} \mathcal{D}_t \leq e^{-\lambda t} \mathcal{D}_0.$$

In particular,

$$\mathbb{E} W_2^2(\mu_t, \delta_{\bar{X}_t}) \leq e^{-\lambda t} W_2^2(\mu_0, \delta_{\bar{X}_0}).$$

Thus the transported measure μ_t concentrates around its random center of mass $\bar{X}_t$.

Proof. Fix two labels $u, v \in \mathbb{R}^d$. Define the difference

$$Z_t^{u,v} := X_t(u) - X_t(v).$$

We first prove that under Assumption 4

$$\mathbb{E}|Z_t^{u,v}|^2 \leq e^{-\lambda t}|u - v|^2,$$

then we integrate this estimate with respect to $\mu_0(du)\mu_0(dv)$.

Writing equation (23) for $X_t(u)$ and for $X_t(v)$ and then subtracting the two equations gives

$$dZ_t^{u,v} = G_t^{u,v} dt + \sum_{\ell=1}^m H_{\ell,t}^{u,v} dB_t^\ell,$$

where the drift coefficient

$$\begin{aligned} G_t^{u,v} &:= b(X_t(u), \Theta_t) - b(X_t(v), \Theta_t) \\ &\quad + \int_{\mathbb{R}^d} [K(X_t(u), X_t(w), \Theta_t) - K(X_t(v), X_t(w), \Theta_t)] \mu_0(dw), \end{aligned}$$

and the diffusion coefficient is

$$H_{\ell,t}^{u,v} := \sigma_\ell(X_t(u), \Theta_t) - \sigma_\ell(X_t(v), \Theta_t).$$

Since $\mu_t = \mu_0 \circ X_t^{-1}$, we may rewrite the interaction term as

$$\int_{\mathbb{R}^d} K(X_t(u), X_t(w), \Theta_t) \mu_0(dw) = \int_{\mathbb{R}^d} K(X_t(u), z, \Theta_t) \mu_t(dz).$$

Thus

$$\begin{aligned} G_t^{u,v} &:= b(X_t(u), \Theta_t) - b(X_t(v), \Theta_t) \\ &\quad + \int_{\mathbb{R}^d} [K(X_t(u), z, \Theta_t) - K(X_t(v), z, \Theta_t)] \mu_t(dz). \end{aligned}$$

Applying Itô's formula to $|Z_t^{u,v}|^2$, we obtain

$$\begin{aligned} d|Z_t^{u,v}|^2 &= 2\langle Z_t^{u,v}, G_t^{u,v} \rangle dt + \sum_{\ell=1}^m |H_{\ell,t}^{u,v}|^2 dt \\ &\quad + 2 \sum_{\ell=1}^m \langle Z_t^{u,v}, H_{\ell,t}^{u,v} \rangle dB_t^\ell. \end{aligned}$$

Apply Assumption 4 with $x = X_t(u)$, $y = X_t(v)$, $\theta = \Theta_t$, and $\nu = \mu_t$. Then

$$2\langle Z_t^{u,v}, G_t^{u,v} \rangle + \sum_{\ell=1}^m |H_{\ell,t}^{u,v}|^2 \leq -\lambda |Z_t^{u,v}|^2.$$

Therefore

$$(24) \quad d|Z_t^{u,v}|^2 \leq -\lambda |Z_t^{u,v}|^2 dt + 2 \sum_{\ell=1}^m \langle Z_t^{u,v}, H_{\ell,t}^{u,v} \rangle dB_t^\ell.$$

Set the new variable

$$Y_t^{u,v} := e^{\lambda t} |Z_t^{u,v}|^2.$$

By Itô's formula,

$$\begin{aligned} dY_t^{u,v} &= e^{\lambda t} \left[\lambda |Z_t^{u,v}|^2 + 2\langle Z_t^{u,v}, G_t^{u,v} \rangle + \sum_{\ell=1}^m |H_{\ell,t}^{u,v}|^2 \right] dt \\ &\quad + 2e^{\lambda t} \sum_{\ell=1}^m \langle Z_t^{u,v}, H_{\ell,t}^{u,v} \rangle dB_t^\ell. \end{aligned}$$

By the drift bound equation (24), the drift term above is nonpositive. Hence

$$dY_t^{u,v} \leq 2e^{\lambda t} \sum_{\ell=1}^m \langle Z_t^{u,v}, H_{\ell,t}^{u,v} \rangle dB_t^\ell.$$

To make this completely rigorous, introduce the localization sequence

$$\tau_R := \inf\{t \geq 0 : |X_t(u)| + |X_t(v)| + |\Theta_t| \geq R\}.$$

Since the solution is global and continuous, $\tau_R \uparrow \infty$ almost surely as $R \rightarrow \infty$. Stopping this inequality at τ_R , we get

$$Y_{t \wedge \tau_R}^{u,v} \leq Y_0^{u,v} + 2 \sum_{\ell=1}^m \int_0^{t \wedge \tau_R} e^{\lambda s} \langle Z_s^{u,v}, H_{\ell,s}^{u,v} \rangle dB_s^\ell.$$

The stopped stochastic integral is a true martingale. Therefore, taking expectations gives

$$\mathbb{E}Y_{t \wedge \tau_R}^{u,v} \leq Y_0^{u,v}.$$

Since $Y_0^{u,v} = |u - v|^2$, we have

$$\mathbb{E} \left[e^{\lambda(t \wedge \tau_R)} |Z_{t \wedge \tau_R}^{u,v}|^2 \right] \leq |u - v|^2.$$

Letting $R \rightarrow \infty$ and using Fatou's lemma yields

$$\mathbb{E} \left[e^{\lambda t} |Z_t^{u,v}|^2 \right] \leq |u - v|^2.$$

Therefore

$$\mathbb{E}|X_t(u) - X_t(v)|^2 \leq e^{-\lambda t} |u - v|^2.$$

This proves the pairwise synchronization estimate.

Now integrate the pairwise synchronization estimate with respect to $\mu_0(du)\mu_0(dv)$. Since $\mu_0 \in \mathcal{P}_2(\mathbb{R}^d)$,

$$\mathcal{D}_0 = \iint |u - v|^2 \mu_0(du)\mu_0(dv) < \infty.$$

By Fubini's theorem,

$$\begin{aligned} \mathbb{E}\mathcal{D}_t &= \mathbb{E} \iint |X_t(u) - X_t(v)|^2 \mu_0(du)\mu_0(dv) \\ &= \iint \mathbb{E}|X_t(u) - X_t(v)|^2 \mu_0(du)\mu_0(dv) \\ &\leq e^{-\lambda t} \iint |u - v|^2 \mu_0(du)\mu_0(dv). \end{aligned}$$

Hence $\mathbb{E}\mathcal{D}_t \leq e^{-\lambda t} \mathcal{D}_0$, proving the first estimate in the theorem.

Therefore

$$\mathbb{E}\mathcal{V}_t = \frac{1}{2} \mathbb{E}\mathcal{D}_t \leq \frac{1}{2} e^{-\lambda t} \mathcal{D}_0 = e^{-\lambda t} \mathcal{V}_0.$$

This proves the internal fluctuation decay estimate.

Finally, since $\mu_t = \mu_0 \circ X_t^{-1}$,

$$\int_{\mathbb{R}^d} |x - \bar{X}_t|^2 \mu_t(dx) = \int_{\mathbb{R}^d} |X_t(u) - \bar{X}_t|^2 \mu_0(du) = \mathcal{V}_t.$$

For the Dirac measure $\delta_{\bar{X}_t}$,

$$W_2^2(\mu_t, \delta_{\bar{X}_t}) = \int_{\mathbb{R}^d} |x - \bar{X}_t|^2 \mu_t(dx).$$

Therefore

$$W_2^2(\mu_t, \delta_{\bar{X}_t}) = \mathcal{V}_t.$$

Taking expectations and using the previous estimate gives

$$\mathbb{E}W_2^2(\mu_t, \delta_{\bar{X}_t}) \leq e^{-\lambda t} W_2^2(\mu_0, \delta_{\bar{X}_0}).$$

□

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